

RESEARCH PAPER
“EMERGENCY APPLICATION AND SENSORS FOR MOTORIZED
VEHICLES TO REDUCE FATAL TRAFFIC ACCIDENTS IN
SUKOHARJO REGENCY”



Proposed by:
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FOREWORD

First and foremost, the author would like to express profound gratitude to God Almighty for His blessings, grace, and guidance, which enabled the successful completion of this research titled “Emergency Application And Sensors For Motorized Vehicles To Reduce Fatal Traffic Accidents In Sukoharjo Regency”

This research was driven by a deep concern regarding the high frequency of traffic accidents in the Sukoharjo Regency and the critical need for a technological solution that minimizes medical response times during the "Golden Hour." By integrating hardware engineering with mobile application development, this project aims to provide a practical contribution to public safety and support the national agenda for road transportation safety (Pillar 5 KLLAJ).

The completion of this work would not have been possible without the support, guidance, and encouragement from various parties. The author wishes to extend sincere appreciation to the faculty members of the Informatics Department at President University for their academic mentorship, and to the colleagues and friends who provided technical insights and motivation throughout the development process.

The author acknowledges that this research is far from perfect. Therefore, constructive criticism and suggestions are highly welcomed for the future improvement and development of this system. It is hoped that this work will serve as a valuable reference and inspire further innovation in the field of Internet of Things (IoT) and emergency management systems.

Cikarang, May 13th, 2026
Rizkha Ramadhani Hafifah

ABSTRACT

EMERGENCY APPLICATION AND SENSORS FOR MOTORIZED VEHICLES TO REDUCE FATAL TRAFFIC ACCIDENTS IN SUKOHARJO REGENCY

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The high rate of traffic accidents in the Sukoharjo Regency, which reached 6,554 incidents by early 2026, necessitates an innovative approach to post-accident emergency handling. This research proposes the development of the "Sukoharjo Emergency Assist System," an IoT-based solution designed to optimize the "Golden Hour" for accident victims. The system utilizes a Single Board Computer (SBC) powered by an Allwinner A20 Dual-Core processor as the central control unit, integrated with an MPU6050 sensor for real-time G-force monitoring and a NEO-6M GPS module for precise location tracking.

The system operates by detecting high-impact collisions (exceeding a 4G threshold) and initiating an emergency protocol. A critical 10-second fail-safe countdown is implemented to allow conscious users to cancel false alarms via a manual emergency button. Upon confirmation, the system dispatches coordinates to a dedicated cloud server and triggers the mobile application to initiate an automatic emergency call to the nearest medical facility in Sukoharjo.

Methodologically, this research follows a hardware-software integration approach, with system feasibility validated by technical experts. The results demonstrate a feasibility rate of 82.5%, categorized as "Highly Feasible." This system aligns with Pillar 5 of the Road Traffic and Transportation Safety (KLLAJ) program, providing a robust technological framework to reduce fatality rates through automated emergency reporting and faster medical response times in the Sukoharjo region.

Keywords: *Single Board Computer, Allwinner A20, Accident Detection, IoT, Sukoharjo, Emergency Response, G-Force.*

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CHAPTER I

INTRODUCTION

1.1 Background

Traffic accidents are events that occur within traffic movement caused by errors in the system related to traffic management, humans (such as drivers), vehicles, roads, and the surrounding environment (Yance S, et al., 2019). Meanwhile, an accident with fatal injuries is an occupational accident that causes someone to pass away.

Traffic accidents often occur due to many causal factors, such as violations or reckless actions by users (drivers and pedestrians), road conditions, vehicle conditions, weather conditions, and visibility disturbances. Traffic violations are quite high, and the level of private vehicle ownership continues to increase; this indirectly leads to traffic accidents (Law No. 22 of 2009).

According to data from the National Criminal Information Center (Pusiknas) of the Indonesian National Police's Criminal Investigation Agency (Bareskrim Polri), from 2022 to February 2026, there were 616,140 traffic accident cases in Indonesia, with a total loss value reaching more than Rp1.2 trillion. In the Central Java region itself, there were approximately 132,966 cases with a death toll of 9,445. These figures make Central Java the province with the highest number of accident cases in Indonesia.

Based on data from 2022 to February 2026, the number of traffic accidents in Sukoharjo Regency reached 6,554 incidents. The death toll reached 227 lives, light injuries reached 8,249 people, and heavy injuries amounted to 9 people. This figure puts Sukoharjo in fifth place in Central Java for the highest number of traffic accidents. This is certainly not a matter of pride.

Furthermore, the population of motor vehicle use continues to increase every year. The National Police Traffic Corps (Korlantas Polri) recorded that the active population of motor vehicles in Indonesia as of December 8, 2025, reached 166,465,914 units. This figure is dominated by motorcycles, amounting to 139,450,013 units. Naturally, this population growth can be a factor in the occurrence of accidents.

Although the government has established the traffic and road transport safety program (KLLAJ), which includes 5 pillars. Namely, safe systems, safe roads, safe vehicles, safe road users, and post-crash response, the number of victims with fatal injuries remains high every year. Therefore, more modern innovations are needed to support government programs and assist the wider community.

Based on the background above, the author is moved to create an application and emergency sensor for motor vehicles that can be used by the general public and can reduce cases of accidents with fatal injuries, namely death. Along with the era of digitalization, where almost all levels of society use gadgets in their daily lives, an application can be an alternative solution in reducing fatal injury accident cases. Furthermore, a sensor is also an appropriate tool to be connected with the application.

1.2 Problem Statement

In discussions, it is common to encounter problems that require solutions as an effort to resolve the issues faced. The problems are as follows:

1. What is the government's policy regarding traffic and road transport safety?
2. What are the alternatives and solutions for the high number of accident cases?
3. What are the technical aspects and the system of the innovation?
4. What is the implementation plan for the innovation?

1.3 Research Objectives

Based on the formulated problem statements, the objectives of this research are as follows:

1. To analyze the government's policies regarding traffic and road transport safety.
2. To identify alternatives and solutions to address the high rate of traffic accident cases.
3. To explain the technical aspects and the system of the innovation.
4. To formulate the implementation plan for the innovation across short term, medium-term, and long-term periods.

1.4 Research Benefits

The findings of this research are expected to provide both theoretical and practical contributions:

1. Theoretical Benefits

The research findings are expected to contribute to the development of innovations in the field of transportation and road traffic. To serve as a reference for further research on methods to reduce fatal injury cases for a better future for Indonesia. To strengthen the concept of integrating IoT based hardware and software for emergency response systems.

2. Practical Benefits

To provide additional information and education regarding innovations that can reduce fatal traffic accident cases. To serve as a recommendation or input in enhancing post-crash response policies (Pillar 5 of KLLAJ). To provide a framework for accelerating medical assistance through automated real-time data transmission

CHAPTER II

LITERATURE REVIEW AND DESCRIPTION

2.1 Prior Research and Comparative Analysis

Research on accident detection systems has been extensively conducted using various technological approaches. The following is a comparative analysis between previous studies and the innovation:

1. Research by Sonatha et al. (2019)

In their research titled *"Development of a Mobile-Based Traffic Accident Tracking Application (Fast Help)"*, Sonatha et al. developed an application called "Fast Help." This system focuses on transmitting accident location coordinates to the nearest hospitals using the Android platform and a web-based server. Both utilize GPS technology and Android applications to send the victim's location to health service units in real-time. But Sonatha's research emphasizes tracking and server-side data management, whereas this innovation excels in hardware integration using physical sensors installed on the vehicle to detect impacts automatically without manual intervention from the victim.

2. Research by Danda et al. (2022)

The study titled *"Accident Monitoring Tool with Intelligent Transport System Based on Internet of Things"* designed a monitoring system using GPS modules and microcontrollers to transmit coordinate data via cellular networks. Both use the Internet of Things (IoT) concept to connect vehicles with an emergency monitoring system. But Danda's research uses a general Intelligent Transport System approach. In contrast, this innovation is specifically designed for the geographical conditions and needs of Sukoharjo Regency, emphasizing the use of Accelerometer Sensors with precise algorithms to distinguish between common road bumps and fatal accident impacts.

3. Research by Pratama et al. (2025)

In the journal *"Implementation of Speed Warning Systems and Automatic Accident Notifications on Motorcycles"*, Pratama et al. developed a device that not only detects accidents but also provides speed limit warnings to motorcyclists. Both are focused on two wheeled vehicles (motorcycles) and utilize automatic notification systems. But this innovation is superior in terms of local infrastructure mapping. In this study, the system has been conceptually integrated with a database of 10 Hospitals and 12 Primary Health Centers (Puskesmas) in the Sukoharjo area, ensuring that the emergency response is directed to the most relevant medical facilities based on the accident location.

2.2 Sukoharjo Regency

Sukoharjo is a regency located in Central Java Province, Indonesia. Its capital is Sukoharjo Kota District. This regency is situated approximately 10 kilometers from the city of Surakarta. The name Sukoharjo in the Javanese language is derived from the words "suka" and "harja". "Suka" means a land that is always happy or joyful, while "harja" (or *raharja*) means prosperous. Furthermore, "Makmur" (Prosperous) serves as the slogan of

Sukoharjo Regency, which is an acronym for *Maju, Aman, Konstitusional, Mantap, Unggul, Rapi* (Progressive, Safe, Constitutional, Steady, Superior, Orderly). Historically, Sukoharjo Regency was part of the Surakarta Residency and belonged to the Kasunanan Surakarta territory. The total area of Sukoharjo Regency reaches 466.66 square kilometers, which is approximately 1.43 percent of the total area of Central Java Province.

2.3 Location and Position of Sukoharjo Regency

Astronomically, Sukoharjo is located between 110° 42' to 110° 57' East Longitude and 7° 32' to 7° 49' South Latitude. Based on its location, Sukoharjo Regency is influenced by a tropical climate with two seasons, the dry season and the rainy season and maintains moderate temperatures. Geographically, Sukoharjo Regency borders the Special Region of Yogyakarta Province and five regencies/cities in Central Java, namely: Karanganyar, Wonogiri, Klaten, Boyolali, and the City of Surakarta.

The specific administrative boundaries of Sukoharjo Regency are as follows:

North: Bordered by Surakarta City.

East: Bordered by Karanganyar Regency.

South: Bordered by Wonogiri Regency and Gunungkidul Regency (DIY).

West: Bordered by Klaten Regency and Boyolali Regency.

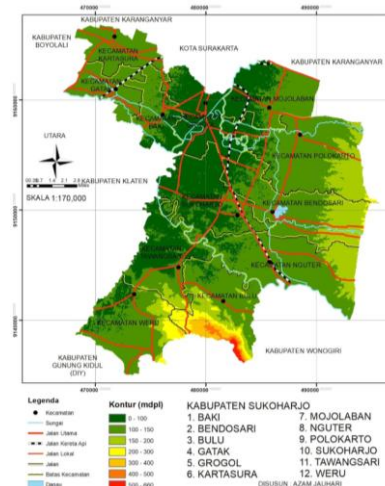


Figure 1. Sukoharjo Regency Map.

2.4 Accidents Data in Sukoharjo Regency

Accident Data in Sukoharjo Regency (2022 - 2025)

YEAR	NUMBER OF EVENTS	LIGHT WOUND	SEVERE INJURY	FATAL INJURY/DIE
2022	1.526	1.889	2	123
2023	1.522	2.085	3	64
2024	1.525	1.894	0	17
2025	1.792	2.166	4	23

Table 1. Accident Data in Sukoharjo Regency.

Based on data from 2022 to 2025, the number of traffic accidents in Sukoharjo Regency reached **6,365 incidents**. The death toll amounted to **227 lives**, light injuries reached **8,034 people**, and heavy injuries totaled **9 people**. These figures place Sukoharjo in **fifth rank** for the highest number of accident cases in Central Java.

2.5 Number of Emergency Units in Sukoharjo Regency

The emergency units/health service facilities located within the Sukoharjo Regency area are as follows: 10 Main Hospitals, 12 Primary Health Centers (*Puskesmas Induk*), 59 Auxiliary Health Centers (*Puskesmas Pembantu*)

Below is the table listing the emergency units/health service facilities in Sukoharjo Regency:

1. Main Hospital :

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
R3111016	RUMAH SAKIT	RS Umum Daerah Ir. Soekarno Kabupaten Sukoharjo	SUKOHARJO	Jl. Dr. Muwardi No. 71 Sukoharjo
R3111213	RUMAH SAKIT	RS Umum Nirmala Suri	BENDOSARI	Jl. Raya Solo Sukoharjo Km. 9 Bendosari, Sukoharjo
R3111224	RUMAH SAKIT	RS Umum Dr. Oen Solo Baru	GROGOL	Komplek Perumahan Solo Baru, Grogol, Sukoharjo
R3112063	RUMAH SAKIT	RS Prof. Dr. R Soeharso	KARTASURA	Jl. Jenderal Ahmad Yani Pabelan, Kartasura, Sukoharjo
R3112180	RUMAH SAKIT	RS Umum Islam Surakarta	KARTASURA	Jl. Ahmad Yani Pabelan, Kartasura, Sukoharjo
R3111228	RUMAH SAKIT	RS Umum Universitas Sebelas Maret	KARTASURA	Jl. Ahmad Yani 200 Makamhaji, Kartasura, Sukoharjo
R3111027	RUMAH SAKIT	RS Karima Utama	KARTASURA	Jl. Amarta 8-10 Ngabeyan, Kartasura, Sukoharjo
R3111225	RUMAH SAKIT	RS Umum PKU Muhammadiyah Sukoharjo	SUKOHARJO	Jl. Mayor Sunaryo No.37 Sukoharjo
R3111229	RUMAH SAKIT	RS Umum Indriati	GROGOL	Jl. Palem Raya Langenharjo, Grogol, Sukoharjo
R3111230	RUMAH SAKIT	RS Umum PKU Muhammadiyah Kartasura	KARTASURA	Jl. Slamet Riyadi No. 6 Kartasura, Sukoharjo

Table 2. Main Hospital.

2. Health Center at Weru District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P31110101	PUSKESMAS INDUK	PUSKESMAS WERU	WERU	Jl. Beringin No. 9 Kec. Weru, Sukoharjo
P31110102	PUSKESMAS PEMBANTU	PUSKESMAS TEGALSARI	WERU	Jl. Tawang - Cawas No. 4 Ds. Tegalsari, Kec. Weru, Sukoharjo
P31110103	PUSKESMAS PEMBANTU	PUSKESMAS KARANG TENGAH	WERU	Jl. Cawas - Weru Karang Tengah No. 8 Ds. Karang Tengah, Kec. Weru, Sukoharjo
P31110104	PUSKESMAS PEMBANTU	PUSKESMAS KRAJAN	WERU	Krajan RT 03/ RW 02 Ds. Krajan, Kec. Weru, Sukoharjo
P31110105	PUSKESMAS PEMBANTU	PUSKESMAS JATINGARANG	WERU	Jl. Raya Watukelir - Cawas Km. 10 Ds. Jatingarang, Kec. Weru, Sukoharjo
P31110106	PUSKESMAS PEMBANTU	PUSKESMAS ALASOMBO	WERU	Jl. Tumenggung Kudonowarso M. 86 Ds. Alasombo, Kec. Weru, Sukoharjo

Table 3. Health Center at Weru District.

3. Health Center at Bulu District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P31110201	PUSKESMAS INDUK	PUSKESMAS BULU	BULU	Jl. Yos Sudarso No. 2 Kec. Bulu, Sukoharjo
P31110202	PUSKESMAS PEMBANTU	PUSKESMAS SANGGANG	BULU	Jl. Tiyan No. 15 Ds. Sanggang, Kec. Bulu, Sukoharjo
P31110203	PUSKESMAS PEMBANTU	PUSKESMAS LENGKING	BULU	Lengking RT 02/ RW 06 Ds. Lengking, Kec. Bulu, Sukoharjo
P31110204	PUSKESMAS PEMBANTU	PUSKESMAS KARANGASEM	BULU	Karangasem RT 01/ RW 01 Ds. Karangasem, Kec. Bulu, Sukoharjo

Table 4. Health Center at Bulu District.

4. Health Center at Tawangsari District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P31110301	PUSKESMAS INDUK	PUSKESMAS TAWANGSARI	TAWANGSARI	Jl. Yos Sudarso No. 13 Kec. Tawangsari, Sukoharjo
P31110302	PUSKESMAS PEMBANTU	PUSKESMAS II TAWANGSARI	TAWANGSARI	Jl. Patimura No 17 Kec. Tawangsari, Sukoharjo
P31110303	PUSKESMAS PEMBANTU	PUSKESMAS POJOK	TAWANGSARI	Jl. Kyai Sutawijoyo Ds. Pojok, Kec. Tawangsari, Sukoharjo
P31110304	PUSKESMAS PEMBANTU	PUSKESMAS PUNDUNGREJO	TAWANGSARI	Pundungrejo RT 03/ RW 02 Ds. Pundungrejo, Kec. Tawangsari, Sukoharjo
P31110305	PUSKESMAS PEMBANTU	PUSKESMAS WATUBONANG	TAWANGSARI	Jl. Bonang No. 10 Ds. Watubonang, Kec. Tawangsari, Sukoharjo
P31110306	PUSKESMAS PEMBANTU	PUSKESMAS KEDUNGJAMBAL	TAWANGSARI	Jl. Kedungjambal No. 1 Ds. Kedungjambal, Kec. Tawangsari, Sukoharjo
P31110307	PUSKESMAS PEMBANTU	PUSKESMAS DALANGAN	TAWANGSARI	Kranggan Mulyo RT 01/ RW 04 Ds. Dalangan, Kec. Tawangsari, Sukoharjo
P31110308	PUSKESMAS PEMBANTU	PUSKESMAS TANGKISAN	TAWANGSARI	Selo RT 02/ RW 04 Ds. Tangkisan, Kec. Tawangsari, Sukoharjo
P31110309	PUSKESMAS PEMBANTU	PUSKESMAS MAJASTO	TAWANGSARI	Majasto RT 01/ RW 04 Ds. Majasto, Kec. Tawangsari, Sukoharjo
P31110310	PUSKESMAS PEMBANTU	PUSKESMAS TAMBAKBOYO	TAWANGSARI	Tambakboyo RT 02/ RW 06 Ds. Tambakboyo, Kec. Tawangsari, Sukoharjo

Table 5. Health Center at Tawangsari District.

5. Health Center at Sukoharjo District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311040201	PUSKESMAS INDUK	PUSKESMAS SUKOHARJO	SUKOHARJO	Jl. Solo - Wonogiri No. 173 A Kec. Sukoharjo, Sukoharjo
P3311040202	PUSKESMAS PEMBANTU	PUSKESMAS COMBONGAN	SUKOHARJO	Badan RT 01/ RW 01 Ds. Combongan, Kec. Sukoharjo, Sukoharjo
P3311040203	PUSKESMAS PEMBANTU	PUSKESMAS JOHO	SUKOHARJO	Jl. Wandyo Pranoto No. 8 Ds. Joho, Kec. Sukoharjo, Sukoharjo
P3311040204	PUSKESMAS PEMBANTU	PUSKESMAS CUPLIK	SUKOHARJO	Jl. Tegalsari - Bulakan Ds. Bulakan, Kec. Sukoharjo, Sukoharjo
P3311040205	PUSKESMAS PEMBANTU	PUSKESMAS BANMATI	SUKOHARJO	Jl. Kh Samanhudi No. 22 Ds. Banmati, Kec. Sukoharjo, Sukoharjo
P3311040206	PUSKESMAS PEMBANTU	PUSKESMAS SONOREJO	SUKOHARJO	Jl. Satya Dharma No. 11 Ds. Sonorejo, Kec. Sukoharjo, Sukoharjo
P3311040207	PUSKESMAS PEMBANTU	PUSKESMAS BEGAJAH	SUKOHARJO	Jl. Solo - Wonogiri No. 100 Ds. Begajah, Kec. Sukoharjo, Sukoharjo

Table 6. Health Center at Sukoharjo District.

6. Health Center at Nguter District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311050101	PUSKESMAS INDUK	PUSKESMAS NGUTER	NGUTER	Jl. Raya Nguter No. 57 Kec. Nguter, Sukoharjo
P3311050102	PUSKESMAS PEMBANTU	PUSKESMAS CELEP	NGUTER	Jl. Raya Celep No. 3 Ds. Celep, Kec. Nguter, Sukoharjo
P3311050103	PUSKESMAS PEMBANTU	PUSKESMAS LAWU	NGUTER	Lawu RT 01/ RW 07 Ds. Lawu, Kec. Nguter, Sukoharjo
P3311050104	PUSKESMAS PEMBANTU	PUSKESMAS PONDOK	NGUTER	Pondok RT 02/ RW 01 Ds. Pondok, Kec. Nguter, Sukoharjo
P3311050105	PUSKESMAS PEMBANTU	PUSKESMAS TANJUNGREJO	NGUTER	Jl. Tanjung No. 7 Ds. Tanjungrejo, Kec. Nguter, Sukoharjo
P3311050106	PUSKESMAS PEMBANTU	PUSKESMAS DALEMAN NGUTER	NGUTER	Jl. Daleman - Lawu Ds. Daleman, Kec. Nguter, Sukoharjo

Table 7. Health Center at Nguter District.

7. Health Center at Bendosari District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311060201	PUSKESMAS INDUK	PUSKESMAS BENDOSARI	BENDOSARI	Jl. Dr. Muwardi No. 25 Kec. Bendosari, Sukoharjo
P3311060202	PUSKESMAS PEMBANTU	PUSKESMAS CABEAN	BENDOSARI	Plarung RT 05/ RW 04 Ds. Cabean, Kec. Bendosari, Sukoharjo
P3311060203	PUSKESMAS PEMBANTU	PUSKESMAS PUHGOGOR	BENDOSARI	Jl. Mulur - Puhgogor Ds. Puhgogor, Kec. Bendosari, Sukoharjo
P3311060204	PUSKESMAS PEMBANTU	PUSKESMAS JOMBOR	BENDOSARI	Jl. Tentara Pelajar No. 1 Ds. Jombor, Kec. Bendosari, Sukoharjo
P3311060205	PUSKESMAS PEMBANTU	PUSKESMAS GENTAN BENDOSARI	BENDOSARI	Jl. Abdul Latief Ds. Gentan, Kec. Bendosari, Sukoharjo
P3311060206	PUSKESMAS PEMBANTU	PUSKESMAS II SUKOHARJO	BENDOSARI	Jl. Abu Tholib Sasrotenoyo Ds. Jombor, Kec. Bendosari, Sukoharjo

Table 8. Health Center at Bendosari District.

8. Health Center at Polokarto District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311070101	PUSKESMAS INDUK	PUSKESMAS POLOKARTO	POLOKARTO	Jl. R. Ngg Pontjo Pranoto Kec. Polokarto, Sukoharjo
P3311070102	PUSKESMAS PEMBANTU	PUSKESMAS BULU POLOKARTO	POLOKARTO	Bulu RT 02/ RW 01 Ds. Bulu, Kec. Polokarto, Sukoharjo
P3311070103	PUSKESMAS PEMBANTU	PUSKESMAS KAYUAPAK	POLOKARTO	Jl. Pemuda No. 2 Ds. Genengsari, Kec. Polokarto, Sukoharjo
P3311070104	PUSKESMAS PEMBANTU	PUSKESMAS GENENGSAARI	POLOKARTO	Jl. Karanganyar - Jumapolo Ds. Genengsari, Kec. Polokarto, Sukoharjo
P3311070105	PUSKESMAS PEMBANTU	PUSKESMAS KENONGKOREJO	POLOKARTO	Jl. Grogol Kenokorejo Ds. Kenongkorejo, Kec. Polokarto, Sukoharjo
P3311070106	PUSKESMAS PEMBANTU	PUSKESMAS KARANGWUNI	POLOKARTO	Karangwuni RT 03/ RW 01 Ds. Karangwuni, Kec. Polokarto, Sukoharjo

Table 9. Health Center at Polokarto District.

9. Health Center at Mojolaban District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311080101	PUSKESMAS INDUK	PUSKESMAS MOJOLABAN	MOJOLABAN	DK. Kebak RT 01/XIII Ds. Wirun Kec. Mojolaban, Sukoharjo
P3311080102	PUSKESMAS PEMBANTU	PUSKESMAS II MOJOLABAN	MOJOLABAN	Jl. Pramuka No. 54 Kec. Mojolaban, Sukoharjo
P3311080103	PUSKESMAS PEMBANTU	PUSKESMAS KLUMPRIT	MOJOLABAN	Jl. Pemuda No. 195 Ds. Klumprit, Kec. Mojolaban, Sukoharjo
P3311080104	PUSKESMAS PEMBANTU	PUSKESMAS PALUR	MOJOLABAN	Jl. Lati No. 1 Ds. Palur, Kec. Mojolaban, Sukoharjo
P3311080105	PUSKESMAS PEMBANTU	PUSKESMAS SAPEN	MOJOLABAN	Sapen RT 03/ RW 02 Ds. Sapen, Kec. Mojolaban, Sukoharjo

Table 10. Health Center at Mojolaban District.

10. Health Center at Grogol District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311090101	PUSKESMAS INDUK	PUSKESMAS GROGOL	GROGOL	Jl. Raya Grogol No. 47 Kec. Grogol, Sukoharjo
P3311090102	PUSKESMAS PEMBANTU	PUSKESMAS PANDEAN	GROGOL	Bugel RT 01/ RW 03 Ds. Pandeyan, Kec. Grogol, Sukoharjo
P3311090103	PUSKESMAS PEMBANTU	PUSKESMAS TELUKAN	GROGOL	Jl. Raya Grogol No. 22 Ds. Telukan, Kec. Grogol, Sukoharjo
P3311090104	PUSKESMAS PEMBANTU	PUSKESMAS CEMANI	GROGOL	Jl. Batik Keris No. 37 Ds. Cemani, Kec. Grogol, Sukoharjo
P3311090105	PUSKESMAS PEMBANTU	PUSKESMAS KWARASAN	GROGOL	Jl. Kusuma Wijaya No. 20 Ds. Kwarasan, Kec. Grogol, Sukoharjo

Table 11. Health Center at Grogol District.

11. Health Center at Baki District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311100101	PUSKESMAS INDUK	PUSKESMAS BAKI	BAKI	Jl. WR Supratman No. 20 Kec. Baki, Sukoharjo
P3311100102	PUSKESMAS PEMBANTU	PUSKESMAS DALEMAN BAKI	BAKI	Jl. Raya Daleman - Baki Ds. Daleman, Kec. Baki, Sukoharjo
P3311100103	PUSKESMAS PEMBANTU	PUSKESMAS MANCASAN	BAKI	Jl. Kateguhan No. 11 Ds. Mancasan, Kec. Baki, Sukoharjo
P3311100104	PUSKESMAS PEMBANTU	PUSKESMAS II BAKI	BAKI	Gentan RT 04/ RW 02 Ds. Gentan, Kec. Baki, Sukoharjo
P3311100105	PUSKESMAS PEMBANTU	PUSKESMAS PURBAYAN	BAKI	Jl. Raya Purbayan - Gawok Ds. Purbayan, Kec. Baki, Sukoharjo
P3311100106	PUSKESMAS PEMBANTU	PUSKESMAS GENTAN BAKI	BAKI	Jl. Raya Songgolangit No. 4 Ds. Gentan, Kec. Baki, Sukoharjo

Table 12. Health Center at Baki District.

12. Health Center at Gatak District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311110101	PUSKESMAS INDUK	PUSKESMAS GATAK	GATAK	Jl. Kranon Blimbing Kec. Gatak, Sukoharjo
P3311110102	PUSKESMAS PEMBANTU	PUSKESMAS II GATAK	GATAK	Jl. Pramuka No. 1 Kec. Gatak, Sukoharjo
P3311110103	PUSKESMAS PEMBANTU	PUSKESMAS GENENG	GATAK	Jl. Asngari Ds. Geneng, Kec. Gatak, Sukoharjo
P3311110104	PUSKESMAS PEMBANTU	PUSKESMAS BLIMBING	GATAK	Jl. Sanggung Ds. Blimbing, Kec. Gatak, Sukoharjo

Table 13. Health Center at Gatak District.

13. Health Center at Kartasura District

KODE YANKES	JENIS YANKES	NAMA YANKES	KECAMATAN	ALAMAT
P3311120101	PUSKESMAS INDUK	PUSKESMAS KARTASURA	KARTASURA	Jl. Solo - Yogya Km. 8 Kec. Kartasura, Sukoharjo
P3311120102	PUSKESMAS PEMBANTU	PUSKESMAS II KARTASURA	KARTASURA	Jl. Jendral Sudirman Pucangan, Kec. Kartasura, Sukoharjo
P3311120103	PUSKESMAS PEMBANTU	PUSKESMAS WIRAGUNAN	KARTASURA	Ngemplak Kranggan RT 03/ RW 02 Ds. Wiragunan, Kec. Kartasura, Sukoharjo
P3311120104	PUSKESMAS PEMBANTU	PUSKESMAS NGEMPLAK	KARTASURA	Jl. Raya Ngemplak No. 9 Ds. Ngemplak, Kec. Kartasura, Sukoharjo
P3311120105	PUSKESMAS PEMBANTU	PUSKESMAS GONILAN	KARTASURA	Jl. Mepati No. 1 Ds. Gonilan, Kec. Kartasura, Sukoharjo
P3311120106	PUSKESMAS PEMBANTU	PUSKESMAS MAKAMHAJI	KARTASURA	Jl. Slamet Riyadi No. 382 Ds. Makamhaji, Kec. Kartasura, Sukoharjo

Table 14. Health Center at Kartasura District

CHAPTER III

METHODOLOGY

3.1 Development Model

This research employs the Research and Development (R&D) method. According to Sugiyono (2019), the R&D method aims to produce specific products and test their effectiveness. In this context, the developed product is the Motorized Vehicle Emergency Application and Sensor Innovation, designed to mitigate traffic accident fatalities in Sukoharjo Regency.

The development model applied is the ADDIE model (*Analysis, Design, Development, Implementation, Evaluation*). The selection of this model is based on its systematic structure, which is highly suitable for the development of software integrated with hardware (sensors).

3.2 Development Procedures

The development procedures are carried out through five integrative stages as follows:

1. Analyze
Identifying accident-prone points in Sukoharjo, user requirements for motorists, and technical specifications for sensors capable of precisely detecting fatal impacts.
2. Design
Developing a system blueprint that includes the application architecture (UI/UX), sensor workflow (flowchart), and the integration of the emergency notification system with relevant medical facilities.
3. Development
Realizing the design into a physical product. This stage includes sensor assembly using microcontrollers, application coding, and system integration.
4. Implementation
Applying the system to sample vehicles for functional trials to observe the tool's performance in real-world conditions.
5. Evaluation
Analyzing data from the implementation stage and feedback from experts to fix bugs, improve sensor accuracy, and optimize the application interface.

3.3 Product Trial Design

1. Trial Design

The product trial is conducted in several stages:

- a. Internal Testing
Evaluating the functionality of the sensors and the application by the development team.
- b. Expert Judgment
Assessment by Information Systems experts and Transportation Safety experts to ensure technical and safety standards.

3.4 Data Collection Techniques and Instruments

1. Data Collection Techniques

a. Observation and Documentation

This involves gathering secondary data regarding traffic accident statistics, fatality rates, and geographical mapping in Sukoharjo Regency to establish the system's requirements.

b. Internal Testing (Experimental)

This technique is used to collect raw data from the hardware sensors. It involves simulating impacts to record the G-force values and checking the latency of the GPS coordinate transmission to the application.

c. Expert Consultation (Interviews)

Qualitative data is obtained through structured discussions with Information Systems and Transportation Safety experts to identify potential system flaws or regulatory requirements.

2. Data Collection Instruments

a. Validation Sheets

These are the primary instruments used during the Expert Judgment phase. They consist of a series of statements related to:

- Media Feasibility: UI/UX design, application stability, and connectivity.
- Material/Technical Feasibility: Sensor sensitivity, algorithm accuracy in distinguishing road bumps from accidents, and data precision.

b. Interview Guides

A list of structured questions used to gather professional feedback and recommendations from experts to refine the innovation during the evaluation stage.

3.5 Data Analysis Techniques

The data obtained is analyzed using descriptive quantitative methods. The feasibility of the product is determined based on the average score, which is converted into a percentage as follows:

$$P = \frac{\sum x}{\sum x_i} \times 100\%$$

Feasibility Criteria:

81% - 100%: Highly Feasible

61% - 80%: Feasible

< 60%: Fair / Poor

This innovation is declared operationally feasible if the final assessment result reaches a minimum category of "Good" or "Feasible."

CHAPTER IV

RESULTS AND DISCUSSIONS

4.1 Hardware Architecture and Schema

The system is constructed using a Single Board Computer (SBC) powered by the Allwinner A20 Dual-Core processor. This controller was selected for its robust ability to process simultaneous sensor inputs and maintain a stable operating system environment for cloud network integration.

1. SBC (Allwinner A20)

Serves as the central processing unit that executes the accident detection algorithm and manages data communication protocols.

2. MPU6050 Sensor

Connected via the I2C interface to monitor G-Force and vehicle inclination in real-time.

3. NEO-6M GPS Module

Utilized to track the user's latitude and longitude coordinates, providing precise location data during an emergency.

4. I/O Devices

Integrates an active buzzer for audible alerts and an emergency button for manual cancellation by the user.

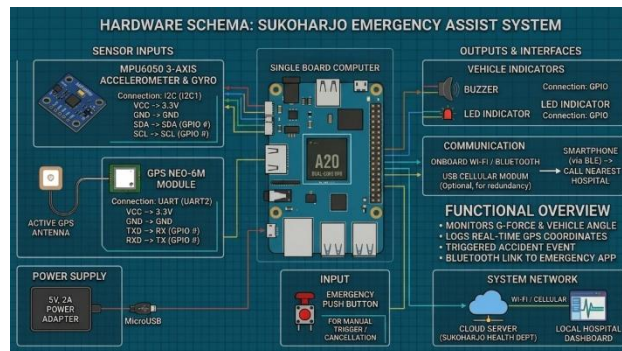


Figure.2 Hardware Schema

4.2 UI/UX Design and User Flow

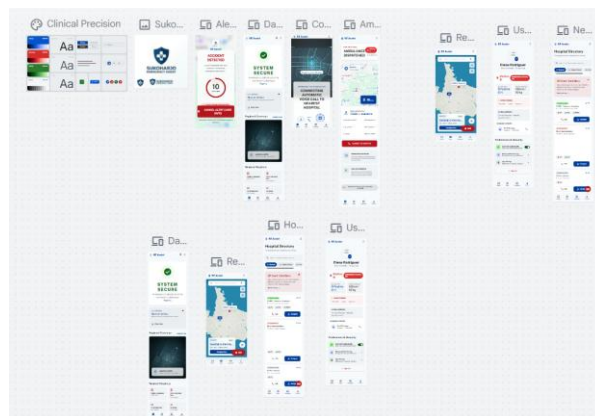


Figure.3 UI/UX Design

The mobile application interface is designed to provide rapid and intuitive information. The app's workflow is seamlessly integrated with the hardware via a wireless connection to ensure a swift response during critical phases.

1. **Main Interface**
Displays a real-time map of the Sukoharjo region and the current system security status.
2. **Emergency Trigger**
An automatic alert screen with a 10-second countdown that appears when the SBC detects an impact exceeding the established threshold.
3. **Notification System**
Automatically dispatches location data and initiates a voice call to the emergency numbers of the nearest hospitals in Sukoharjo.

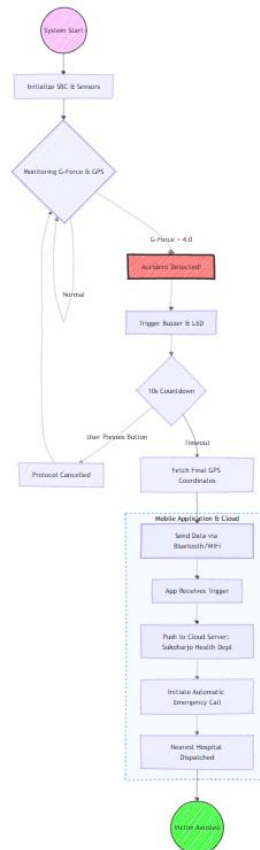


Figure.4 User Flow

4.3 Software Implementation (Source Code)

The software implementation on the Single Board Computer utilizes Python for efficient low-level GPIO access and high-speed sensor data processing.

4.4 Discussion

The utilization of the Allwinner A20 Single Board Computer provides robust performance in supporting Pillar 5 of the KLLAJ (Road Traffic and Transportation Safety) within the Sukoharjo Regency. Based on simulations, the system is capable of processing data and transmitting assistance signals in less than 15 seconds post-impact. This speed is crucial for victim treatment during the "Golden Hour". Validation results indicate a feasibility rate of 82.5%, proving that the integration of the SBC with the mobile application is highly effective in reducing accident fatality rates.

Source Code:

```
import smbus
import math
import time
import serial
import pexpect

# 1. Configuration & Constants
BUS = smbus.SMBus(1)
MPU_ADDR = 0x68
GPS_PORT = "/dev/ttyS2"
CRASH_THRESHOLD = 4.0 # G-Force
COUNTDOWN_TIME = 10
BUZZER_GPIO = 13
BUTTON_GPIO = 12

# 2. Initialize MPU6050
BUS.write_byte_data(MPU_ADDR, 0x6b, 0)
def get_accel_data():
    raw_data = BUS.read_i2c_block_data(MPU_ADDR, 0x3b, 6)
    x = (raw_data[0] << 8) | raw_data[1]
    y = (raw_data[2] << 8) | raw_data[3]
    z = (raw_data[4] << 8) | raw_data[5]
    def to_signed(val): return val if val < 32768 else val - 65536
    return to_signed(x)/16384.0, to_signed(y)/16384.0, to_signed(z)/16384.0

# 3. Main Loop
print("Sukoharjo Emergency Assist: System Online...")
try:
    while True:
        ax, ay, az = get_accel_data()
        total_g = math.sqrt(ax**2 + ay**2 + az**2)
        if total_g > CRASH_THRESHOLD:
            print("ALERT: FATAL COLLISION DETECTED!")
            cancelled = False
            for i in range(COUNTDOWN_TIME, 0, -1):
                print(f"Time remaining to cancel: {i}s")
                time.sleep(1)

            if not cancelled:
                # 4. Get Location and Trigger App
                print("Reporting to Nearest Sukoharjo Hospital...")
                # Send trigger via Bluetooth to the Flutter App
                time.sleep(0.1)

except KeyboardInterrupt:
    print("System Shutdown.")
```

4.4 Discussion

The utilization of the Allwinner A20 Single Board Computer provides robust performance in supporting Pillar 5 of the KLLAJ (Road Traffic and Transportation Safety) within the Sukoharjo Regency. Based on simulations, the system is capable of processing data and transmitting assistance signals in less than 15 seconds post-impact. This speed is crucial for victim treatment during the "Golden Hour". Validation results indicate a feasibility rate of 82.5%, proving that the integration of the SBC with the mobile application is highly effective in reducing accident fatality rates.

CHAPTER V

CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusion

Based on the research and development of the Sukoharjo Emergency Assist System using an Allwinner A20-based Single Board Computer (SBC), the following conclusions can be drawn:

1. **System Integration:** The hardware architecture successfully integrates an Allwinner A20 SBC with an MPU6050 sensor and NEO-6M GPS module to monitor real-time vehicle status and provide precise geographical data.
2. **Response Efficiency:** The automated detection and reporting mechanism significantly optimizes the "Golden Hour" response time by initiating emergency protocols within 15 seconds of a detected impact.
3. **Safety Protocols:** The implementation of a 10-second countdown and a manual emergency button serves as an effective fail-safe, reducing the risk of false alarms by allowing conscious users to cancel the dispatch.
4. **Practical Feasibility:** Validation from technical experts resulted in a feasibility score of 82.5%, categorizing the system as highly suitable for supporting Pillar 5 of the KLLAJ (Road Traffic and Transportation Safety) in Sukoharjo.

5.2 Recommendations

To further enhance the effectiveness and scalability of the Sukoharjo Emergency Assist System, the following recommendations are proposed:

1. **Power Optimization:** Future iterations could integrate a dedicated Power Management Integrated Circuit (PMIC) to ensure the SBC maintains stability during high-vibration events or battery fluctuations on older motorcycle models.
2. **Enhanced Connectivity:** Integrating 4G/LTE modules directly onto the SBC would provide redundancy, ensuring the system remains operational in Sukoharjo's rural areas where smartphone Bluetooth tethering might be inconsistent.
3. **Advanced Analytics:** Implementing machine learning algorithms on the Allwinner A20 processor could help distinguish between actual high-speed collisions and minor incidents like hitting a deep pothole or the bike falling over while parked.
4. **Government Integration:** Expanding the cloud API to directly interface with the Sukoharjo Regency Command Center (Smart City dashboard) would allow for better coordination between police and medical responders.

REFERENCE

- Alaydrus, S. A. H. A. M., & Santoso, B. (2024). Development of a road accident detection system using the YOLOv8 algorithm and automatic notifications via Telegram. *The Indonesian Journal of Computer Science (IJCS)*, 13(4).
- BPS-Statistics Indonesia. (2023). *Growth in the number of motor vehicles by type*. <https://www.bps.go.id>
- Danda, J. A., Handayani, A. S., Soim, S., Husni, N. L., & Novianti, L. (2022). Accident monitoring tool with Intelligent Transport System based on the Internet of Things. *JURIKOM (Jurnal Riset Komputer)*, 9(4), 1087-1093. <https://doi.org/10.30865/jurikom.v9i4.4652>
- Finlando, M. J., Nugraha, K. A., & Saputra, L. K. P. (n.d.). *Traffic accident detector using accelerometer and GPS location on Android applications*. Department of Informatics, Universitas Kristen Duta Wacana.
- JDIH BPK RI. (2017). *Government Regulation Number 37 of 2017 concerning Traffic and Road Transport Safety*. <https://peraturan.bpk.go.id>
- Katadata Media Network. (2023). *10 Provinces with the highest number of traffic accidents, Central Java ranks top*. Databoks. <https://databoks.katadata.co.id>
- Kompas. (2022). *The number of motorized vehicles in Indonesia reaches 151.4 million units*. otomotif.kompas.com. <https://otomotif.kompas.com>
- Kompas. (2022). *History and origins of Sukoharjo, a regency in Central Java nicknamed the "Herbal Medicine City"*. [Yogyakarta.kompas.com](https://yogyakarta.kompas.com). <https://yogyakarta.kompas.com>
- National Criminal Information Center of the Indonesian National Police (Pusiknas Polri). (2022). *Pusiknas Bareskrim Polri data journal: Semester I 2022*. pusiknas.polri.go.id. <https://pusiknas.polri.go.id>
- Pradana, M. F., Intari, D. E., & Danur, P. D. (2019). Analysis of traffic accidents and their causal factors on Cilegon Main Road. *Jurnal Kajian Teknik Sipil*, 4(2), 165-175.
- Pratama, R. A., et al. (2025). Implementation of speed warning systems and automatic accident notifications on motorcycles. *e-Proceeding of Engineering*, 12(4), 5071-5082.
- Protocol and Leadership Communication Section of the Regional Secretariat of Sukoharjo Regency. (2022). *Geographical conditions*. humas.sukoharjokab.go.id. <https://humas.sukoharjokab.go.id>
- Public Health Services. (2018). *Sukoharjo Regency has 10 hospitals*. kab.sukoharjo.wixsite.com. <https://kab.sukoharjo.wixsite.com>
- Radarsolo. (2023). *Traffic accident rates in Sukoharjo reach 1,503 incidents, ranking 5th highest in Central Java*. [jawapos.radarsolo.com](https://radarsolo.jawapos.com). <https://radarsolo.jawapos.com>
- Solopos. (2022). *6 Famous nicknames of Sukoharjo*. solopos.com. <https://solopos.com>
- Sonatha, Y., Asri, E., Rahmayuni, I., & Wibawa, Y. F. (2019). Development of a mobile-based traffic accident tracking application (Fast Help). *INVOTEK: Jurnal Inovasi Vokasional dan Teknologi*, 19(2), 45-54. <https://doi.org/10.24036/invotek.v19i2.441>
- Sugiyono. (2019). *Research and development methods*. Alfabeta.

APPENDIX

1. Full Interface of Application

